MIL-R-38980A(USAF)
25 March 1970
Supersedes
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12 November 1968

MILITARY SPECIFICATION

RACK, BOMB EJECTOR, AIRCRAFT BRU-4/A AND RACK, BOMB EJECTOR, AIRCRAFT BRU-4A/A

This specification is mandatory for use by USAF.

1. SCOPE

This specification covers the requirements for two types of equipment designated as the Rack, Bomb Ejector, Aircraft BRU-4/A and the Rack, Bomb Ejector, Aircraft BRU-4A/A, hereafter referred to as the rack, for carriage and release (either freefall or forced ejection) of stores up to the 5000 pound class.

2. APPLICABLE DOCUMENTS

The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

2.1 SPECIFICATIONS

Federal

PPP-C-843	Cushioning Material, Cellulose

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-T-7743	Testing, Store Suspension Equipment, General Specification for
MIL-L-10547	Liners, Case, Waterproof
MIL-B-26195	Boxes, Wood-Cleated, Skidded, Load-Bearing Base
MIL-Q-9858	Quality Control System Requirements

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2.2 STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards; Order of Precedence for the Selection of
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-785	Requirements for Reliability Program (For Systems and Equipment)
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods

2.3 DRAWINGS

Air Force

68J13090	Rack, Bomb Ejector, Aircraft BRU-4/A, Assembly of
68J13136	Wire Harness, Rack, Bomb Ejector, Assembly of
69J13400	Rack, Bomb Ejector, Aircraft BRU-4A/A, Assembly of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. This specification makes provisions for preproduction testing.

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3.2 *Components*

3.2.1 Selection of Materials. Specifications and standards of all materials and parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.

3.2.2 Weight Reduction. The contractor shall make every effort to hold to a minimum the equipment's over-all weight without impairing the function, operation or durability. All changes that are required to accomplish weight reduction shall be subject to written approval by the procuring agency prior to incorporation.

3.3 Design and Construction

3.3.1 The racks shall be constructed in conformance to drawing 68J13090-7 (BRU-4/A), 69J13400 (BRU-4A/A), and subsidiary drawings.

3.3.1.1 Precedence. In the event of any conflict between this specification and the drawings, the specification shall govern.

3.3.2 Interchangeability. All parts having the same number shall be directly and completely interchangeable with each other with respect to installation and performance. The racks shall be completely interchangeable from aircraft to aircraft. When parts are not interchangeable, the criteria listed in MIL-D-1000 shall apply and new drawings shall be prepared as required.

3.4 Performance. The racks shall be capable of suspending and releasing stores by means of a cartridge-actuated mechanism which shall allow either a freefall drop or forced ejection of all stores up to the 5000 pound class having 14 or 30 inch lug spacing and a diameter between 10.70 and 33 inches. The rack is designed for dive, horizontal, and other type bombing.

3.4.1 *Forced Ejection. When forced ejection of the store is desired,* the velocity of store, reaction forces and the flight attitude of the store shall be controlled by orifices which are placed in the gas system and which meter the flow of gases to the ejection pistons.

3.4.2 Special Weapon Requirements

3.4.2.1 In-Flight Lock Provision. The racks shall contain a reversible in-flight locking mechanism which, when locked, will prevent weapon release even though the releasing force is generated in the releasing device.

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3.4.2.2 In-Flight Lock Function. With the in-flight lock in the locked position, no electrical energy shall be transmitted to the cartridge when a 28 volt potential is applied to the cartridge firing circuits and the hook release mechanism shall be restricted from operation by a mechanical block.

3.4.3 Locked Shut and Sympathetic Firing. The racks shall be capable of withstanding the stresses incurred when fired with the in-flight safety lock engaged without inadvertently releasing the store or damaging any structural part of the rack. Both cartridges shall fire when only one of the firing switches (S-1, S-3) is closed.

3.4.4 Conventional Weapon Arming Control. The nose, center, and tail solenoid operated arming controls shall operate from an 18 to 30 vdc power supply and shall each draw a maximum of 0.15 amperes continuously when energized at 24 vdc. Each control shall be capable of receiving and holding all standard arming wire loops. Each control shall be selectively capable of providing for an unarmed release or ejection, or armed release or ejection, of stores weighing from 100 to 5000 pounds.

3.4.5 Life. The racks shall function satisfactorily for 250 firings.

3.4.6 Reliability. The contractor shall initiate and maintain a reliability program in accordance with MIL-STD-785. This program shall be subject to modification by the procuring activity.

3.4.6.1 Reliability Levels. The following reliability levels shall be achieved and demonstrated by the contractor:

a. Inadvertent Release. With proper adjustment of the hook release mechanism, there shall be no inadvertent release during the design service life of the racks (250 firings under service conditions).

b. Release Performance Requirements. With proper maintenance and replacement of parts as required per maintenance Tech Order, the achieved rate of failure shall be no failures in 250 releases under service conditions, excluding cartridge failure.

c. Reliability Failure. The bomb rack shall be constructed as specified herein and successfully complete the tests of paragraph 4.5, "Test Methods," of this specification with no failures as defined herein preventing satisfactory performance. Reliability of 98.0 percent for a 2-hour estimated mission time with Mean Time Between Failure (MTBF) value of 100 hours is required. Compliance will be acceptably demonstrated at 90 percent confidence level by 440 hours or more of total qualification testing time with no uncorrected failures.

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3.5 Environmental Requirements. The racks shall be capable of proper operation during or after, as applicable, the following environmental conditions:

a. Temperature from -53.9°C (-65°F) to +176.7°C (350°F).

b. Humidity up to 95 percent, relative.

c. Salt spray incident to use, storage or transportation near or on a large body of salt water.

d. Icing conditions incident to service use.

e. Sand and dust likely to be encountered under desert conditions.

f. Vibration incident to flight, transportation and handling.

g. Shock incident to flight, transportation and handling.

3.6 Identification of Product. The racks and all component parts thereof shall be marked for identification in accordance with MIL-STD-130.

3.7 Government Furnished Property. When specified in the contract, the Government will furnish the following to the contractor, upon his request, for use in testing the Rack, Bomb Ejector, Aircraft BRU-4/A and the Rack, Bomb Ejector, Aircraft BRU-4A/A.

Item Number	Stock Number or Part Number	Description of Item	Quantity
1	1375-516-9924-M187	Cartridge, Bomb, Ejection, CCU-9/B	As stated in contract
2		Cartridge, Bomb Ejection, CCU-10/B	As stated in contract
3		Cartridge, Bomb Ejection, CCU-1/B	As stated in contract

3.8 Workmanship. The racks shall be fabricated and finished in a manner which will insure that the criteria of appearance, fit, smoothness of operation, and adherence to specified tolerances are observed. These criteria must apply to all accessories and to all external nonfunctioning parts, such as nameplates, as well as to functional parts. Particular attention shall be given to neatness, thoroughness of marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

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4. QUALITY ASSURANCE

4.1 Classification of Tests. The inspection and testing of the rack shall be as follows:

a. Preproduction Tests

b. Acceptance Tests

4.2 Preproduction Testing

4.2.1 Preproduction Test Samples. The preproduction test units shall be representative of the production equipment. They shall be tested by a laboratory designated by the procuring activity, or when so stated in the contract, at the contractor's facility under the supervision of the procuring activity.

4.2.2 Preproduction Report. In the event that preproduction testing is performed at the contractor's facility the contractor shall, at the completion of preproduction test, prepare a preproduction test report. The preproduction test report shall be prepared in accordance with MIL-STD-831 and three complete copies shall be furnished to the procuring activity.

4.2.3 Preproduction Tests. Preproduction tests shall consist of all tests described in "4.5 Test Methods" of the specification.

4.3 Acceptance Tests. Acceptance testing shall consist of the following:

a. Individual Tests

b. Sampling Tests

4.3.1 Individual Tests. Each rack shall be subjected to the individual tests as listed in Table I, except as otherwise specified.

4.3.2 Sampling Plan and Tests

4.3.2.1 Sampling Plan. One sample shall be selected at random by the quality assurance representative from each 200 production racks. Prior to being subjected to the sampling tests, the sample racks shall have passed the individual tests.

4.3.2.2 Sampling Tests. Sampling tests shall consist of the sampling tests shown in Table I.

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TABLE I

TEST	REQUIREMENT PARAGRAPH	TEST PARA-GRAPH	QUALIFI-CATION TEST	INDI-VIDUAL TEST	SAMPLING TEST
Examination of Product	3.3, 3.6, 3.8	4.5.1	X	X	
Conformance to Drawings	3.3, 3.8	4.5.2	X	X	
Static Load Tests	3.3	4.5.3	X		X
*Functional Test	3.4	4.5.4	X	X	
**Locked Shut and Sympathetic Firing Test	3.4	4.5.5	X	X	
Reliability Life Test	3.4	4.5.6	X		X
Environmental Tests	3.5	4.5.7			
High Temperature Test	3.5.a	4.5.7.1	X		X
Humidity Test	3.5.b	4.5.7.2	X		
Salt Fog Test	3.5.c	4.5.7.3	X		X
Icing Conditions Test	3.5.d	4.5.7.4	X		X
Sand and Dust Test	3.5.e	4.5.7.5	X		
Vibration Test	3.5.f	4.5.7.6	X		X
Shock Test	3.5.g	4.5.7.7	X		X
Electrical Tests	3.4	4.5.8	X	X	X

* These tests may be performed concurrently with the Reliability Life Test.

**Every twentieth rack shall be tested to this test (Individual tests only).

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4.3.3 Rejection and Retest. When the sample rack fails to meet the specified test, no rack on hand or later produced shall be accepted until the extent and cause of failure are determined. The failure shall be corrected and the unit shall demonstrate compliance by satisfactorily passing the failed test before acceptance of on-hand or later produced units.

4.3.4 Defects in Items Already Accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test Conditions. Test conditions shall be as specified in each specific test under "4.5 Test Methods."

4.5 Test Methods

4.5.1 Examination of Product. Prior to installation of the left side plate each rack shall be examined visually to determine compliance with all requirements of this specification for which tests are not specified. Failure to comply with all requirements shall be cause for rejection.

4.5.2 Conformance to Drawings. Each rack shall be inspected to determine conformance to the engineering drawings, subsidiary drawings and specifications and standards referenced therein. Failure to conform to these criteria shall be cause for rejection.

4.5.3 Static Load Tests

NOTE: Tolerances for all loads applied in the static load tests shall be $\pm$3 percent.

4.5.3.1 Yield Load Tests

4.5.3.1.1 Thirty-Inch Suspension

4.5.3.1.1.1 Test Set-Up. The racks shall be installed in a fixture capable of withstanding the loads specified in the test procedures. This fixture shall be rigidly constructed in a manner such that vertical deflection of the bomb rack shall not exceed 1/2 inch during performance of the firing tests specified in the Functional Test. Pins or bolts for this installation shall have a minimum ultimate tensile strength of 160,000 psi. A 2000 pound simulated store as shown in figure 1 shall be installed on the 30-inch hooks of the rack. The set of swaybraces at each end of the rack shall be torqued to a total of 840 $\pm$ 50 inch pounds, each stud should be at approximately 420 inch pounds. The torque shall be applied incrementally by turning each bolt of a set approximately the same number of turns.

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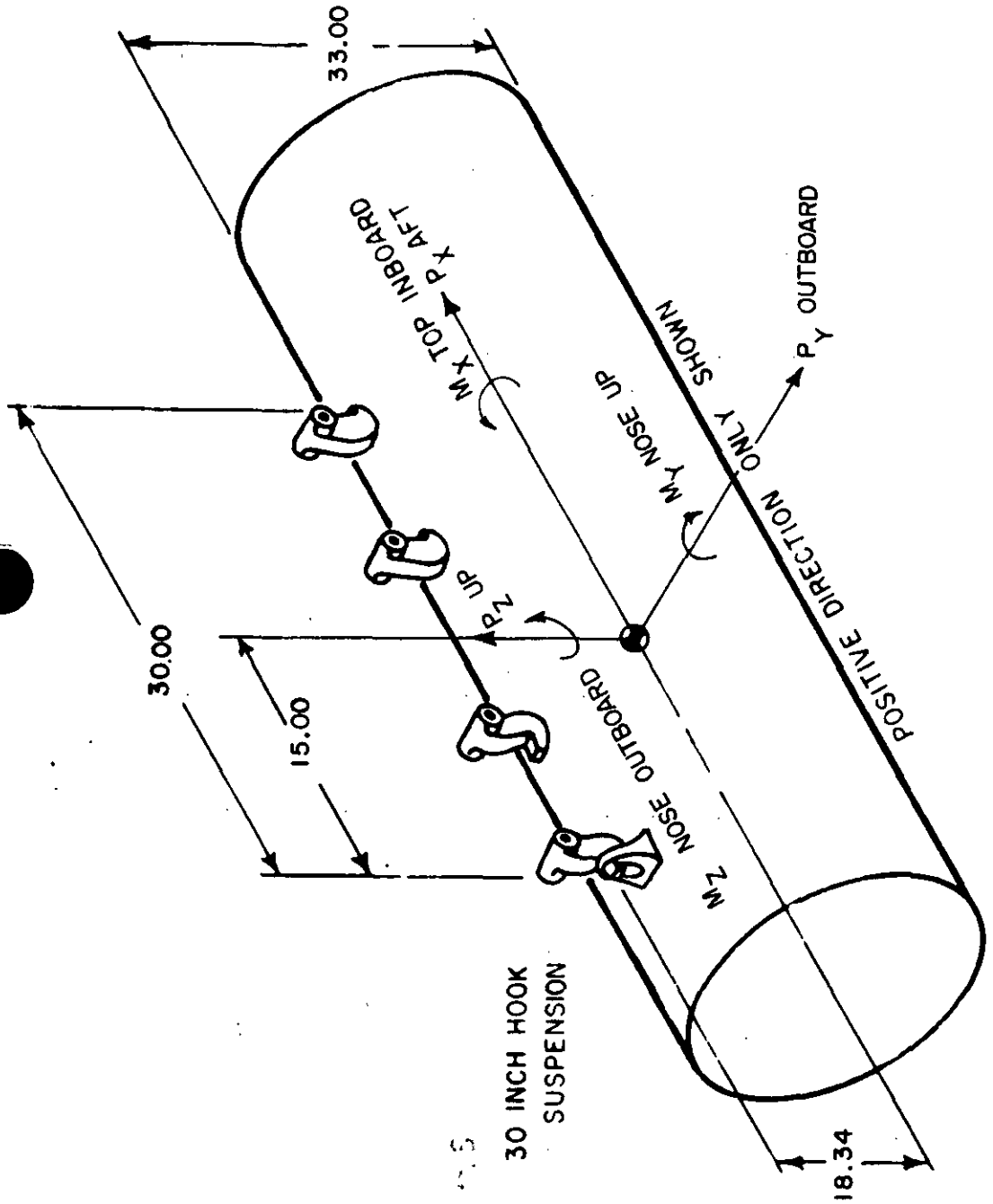


FIGURE 1 LOCATION AND DIRECTION OF LOADS

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4.5.3.1.1.2 Test Procedure. The yield load test of the racks shall consist of the simultaneous imposition of loads for each condition along or about, as applicable, the axes shown in figure 1. For each condition the loads shall first be imposed in increments of 10 percent of design load until 100 percent of design load is reached, then in increments of 5 percent of design load until 115 percent of design load is reached. The loads shall be held at each percent increment for three minutes before proceeding to the next percent increment after 60 percent of the design load is attained. The rate of loading shall not exceed 5 percent of the design loads per second. When a load of 115 percent of design load has been reached and held for three minutes, the load shall be relieved to zero and the rack inspected. The simulated store shall be manually released and the position of the racks in the fixture shall be reversed. Test set-up and procedures shall be repeated. Any permanent deformation shall be cause for rejection. Design loads shall be as shown in Table II.

TABLE II

DESIGN LOADS

	30 INCH HOOKS					
CONDITION NUMBER	P_x Lbs	P_y Lbs	P_z Lbs	M_x In-Lbs	M_y In-Lbs	M_z In-Lbs
1	3050	15600	-12300	36000	292000	-165000
2	3050	10800	-20300	36000	384000	-110000

Reference axes and positive directions of those axes are shown in figure 1.

4.5.3.1.2 Fourteen Inch Suspension

4.5.3.1.2.1 Test Set-Up. Test set-up shall be as specified for the 30-inch suspension except that the simulated store shall be suspended on the 14-inch hooks of the rack as shown in figure 2.

4.5.3.1.2.2 Test Procedure. The test procedure shall be as specified for the 30-inch suspension. Design loads shall be as shown in Table III. The rejection criteria for the 30-inch suspension also applies to the 14-inch suspension.

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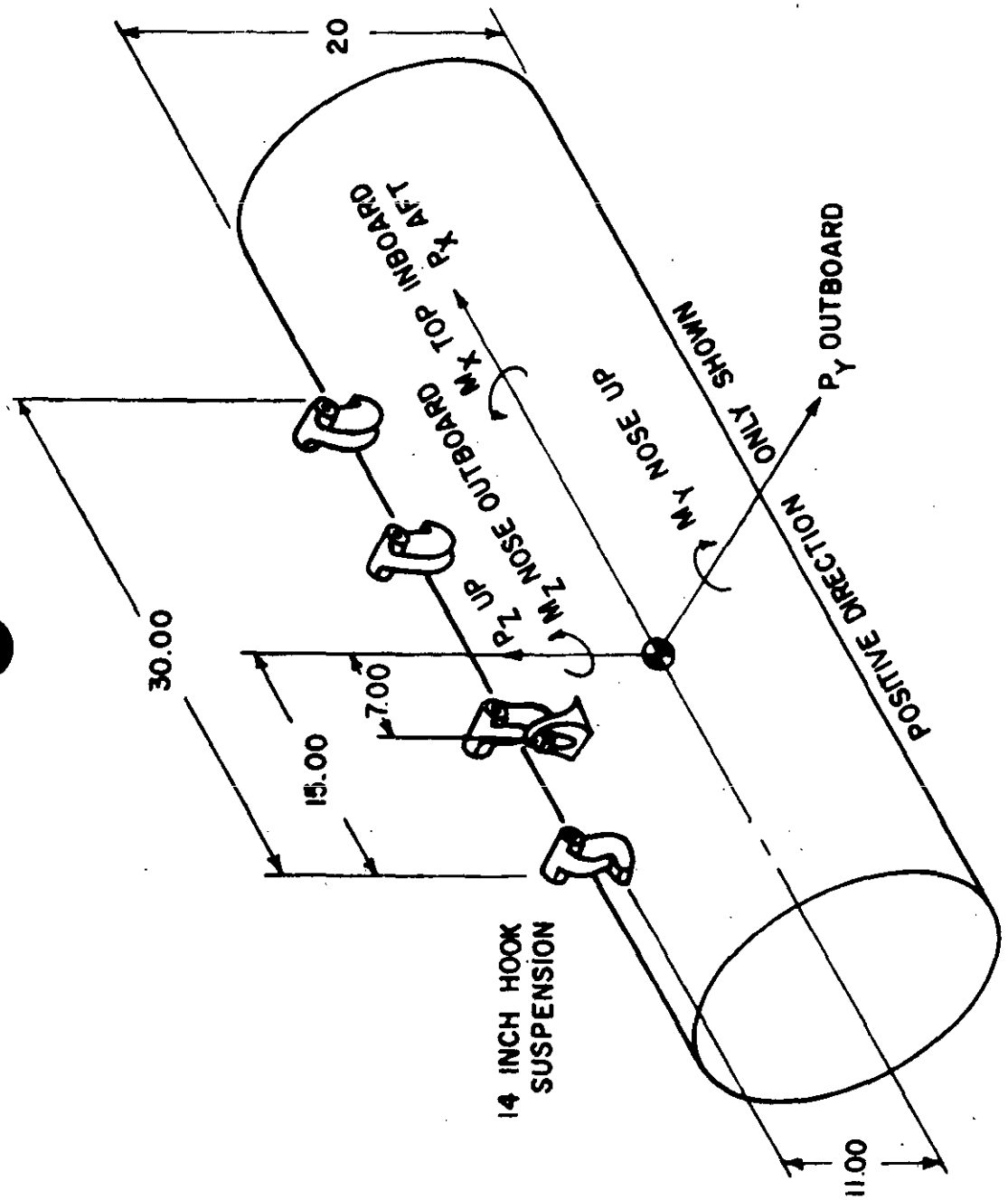


FIGURE 2 LOCATION AND DIRECTION OF LOADS

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TABLE III

	14 INCH HOOKS					
CONDITION NUMBER	P_x Lbs	P_y Lbs	P_Z Lbs	M_x In-Lbs	M_y In-Lbs	M_z In-Lbs
1	350	5610	-3220	10000	-28400	45280
2	240	6740	-3840	15800	-23000	31340
3	350	6975	-4170	16800	-21700	31000
4	670	2935	-4000	0	-125300	83500

4.5.3.2 Ultimate Strength Test

4.5.3.2.1 Thirty Inch Suspension

4.5.3.2.1.1 Test Set-Up. The test set-up for this test shall be as specified for the YIELD LOAD TEST.

4.5.3.2.1.2 Test Procedure. Test procedure shall be as specified for the YIELD LOAD TEST up to 115 percent of the design load, except that after 115 percent of the design load has been reached, loading shall continue in 5 percent increments of the design load up to 150 percent of the design load. After 150 percent of the design load has been reached and held for three minutes, the load shall be reduced to zero and the rack inspected. The racks shall be tested in one direction only. Any breakage during loading up to and including 150 percent of design loads shall be cause for rejection. Design loads shall be as shown in condition 2 only of Table II.

4.5.4 Functional Test. Testing of the racks shall be performed as outlined in the following procedures and in the order listed:

a. The racks shall be mounted on a test stand in a horizontal position with the hooks downward.

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b. A 2000 pound simulated store with standard Air Force Bomb Lugs on 30 inch centers and the center of gravity located midway between the lugs shall be suspended from the racks. The ground safety pin shall be properly inserted. Sway braces shall be torqued as required in the test set-up of the 30 inch suspension yield load test paragraph of this specification. The ground safety pin shall be removed and reinserted. Any movement of the hook linkage or rotation of the center bellcrank that prevents or restricts the removal or reinsertion of the ground safety pin shall be cause for rejection. Two CCU-1/B cartridges shall be inserted in the cartridge retainers (P/N 64D13265) and installed in the breech of the rack. The retainers shall be screwed into the breech with a torque of 250 to 300 inch pounds. Forward and aft orifice pin assemblies with a diameter of 0.111 inch shall be installed. The inflight safety lock shall be unlocked by applying a 28 $\pm$ 4 volt current across pins L and M of the electrical connector. With the ground safety pin removed, 28 $\pm$ 4 volts DC shall be applied to Pins A-j and C-j and the cartridges shall fire. The hooks shall open and the store shall be forcibly ejected. The velocity of the store at the end of the piston stroke shall be 14 $\pm$ 2 feet per second. The ejector pistons shall have returned to their retracted positions. In all cases except individual tests, the peak reaction forces against the supporting structure shall be measured and recorded for each piston. The flight attitude of the store at a position between 2.5 and 3.0 feet below the suspended position shall be horizontal $\pm$ 1-1/2 degrees.

c. The rack hooks shall be latched by clockwise rotation of a suitable tool engaged in the square manual release socket located on the side of the rack containing the cartridge access to the breech and designated by the "Hook Manual Release" presentation. Torque required for this operation shall be 40 $\pm$ 15 inch pounds. At this time the inflight safety lock arm shall be checked through the inspection window in the side of the rack to determine that it is in the fully locked position. This is shown as the "LOCK" position on each side of the rack. If the inflight safety lock does not self-rotate from spring loading to the fully locked position, the rack shall be rejected.

d. With the racks latched and the inflight safety lock arm in locked position as indicated in step c. of this procedure, the manual release tool shall be inserted in the manual release socket and rotation attempted in the direction indicated for manual release. The hooks shall not open.

e. With the racks latched, the inflight safety lock shall be unlocked by inserting a suitable tool in the hexagonal socket located on the unlocked left side of the rack and rotating until the inflight

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safety lock arm is in the unlocked position as shown by the "LOCK-UNLOCK" presentation on the side of the rack. Torque required for this operation shall be 15 ± 5 inch pounds. This operation is shown by the "In-Flight Lock Manual Release" presentation on the side of the rack. The safety lock bell crank shall be pinned to the unlocked position by insertion of 0.248 inch maximum, 0.245 inch minimum, diameter pin through the sideplates and the safety lock switch actuating arm. The hook manual release shall be rotated in a counterclockwise direction. Torque required for this operation shall be 40 ± 15 inch pounds. The hooks shall open and remain in an opened position. All circuits of switches S-1 and S-3 shall be checked for proper continuity in accordance with drawing 68J13136. NOTE: The torque measurements of steps c., d., and e. shall be repeated three times.

f. The rack hooks shall be latched and 28 ± 4 volts DC applied to pins L and M of the electrical plug located at the end of the rack. The inflight safety lock shall open and remain open. Maximum solenoid actuation current shall be 15.5 amps. Maximum holding current shall be 0.175 amp. The hooks shall be manually opened and remain in an opened position.

g. The store away switch shall actuate when the end of the plunger is at a point 1.50 inches minimum below the suspension surface of the 14 inch hooks. The store away switch shall be monitored for proper actuation. Pins S-T, V-W, a-b, d-e shall indicate "store off" and pins T-U, W-X, b-c, e-f shall indicate "store present."

h. A standard arming wire for conventional weapons shall be installed in each of the arming controls. Using a convenient device, the force required to pull the arming wires installed in the arming solenoid shall be measured. This measurement shall be made twice for each arming control, once with the arming solenoid not energized and once with the solenoid energized. The required forces shall be:

(1) Arming solenoid not energized:

The arming control shall not release the arming wire when a force of 10 pounds is applied. The arming control shall release the arming wire when a force of 14 pounds is applied.

(2) Arming solenoid energized:

The arming control shall not release the arming wire when a force of 150 pounds is applied.

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NOTE: Throughout this specification, when orifice diameters are specified, the diameter tolerances shall be those shown in the specific orifice drawing.

4.5.5 Locked Shut and Sympathetic Firing Test. The racks shall be mounted with the top closeout channel removed. The pin connecting S-1 to the safety lock linkage shall be removed through the appropriate hole in the left side plate. A 2000 pound simulated store shall be loaded on the rack. Two CCU-1/B shall be installed and the inflight safety lock engaged. Special overboard orifice assemblies shall be installed with a 0.025 minimum to 0.040 maximum inch diameter. The safety lock bell crank shall remain in the spring-loaded, locked position, S-1 shall be actuated by manually closing its firing circuit. The rack shall be fired. Both cartridges shall have fired. The rack hooks shall not unlatch and the store shall remain suspended on the rack. After the Locked Shut and Sympathetic Firing Test, the rack shall be subjected to steps c., d., and e. of the Functional Test. Failure to meet these criteria or evidence of failure or permanent deformation of any structural part shall be cause for rejection.

4.5.6 Reliability Life Test. Perform step b. of the Functional Test. After completion of step b., of the Functional Test, 100 releases of the 2000 pound simulated store shall be performed at ambient temperature with the fore and aft orifice pin assemblies having 0.111 inch diameters and 100 releases of the 500 pound simulated store with the fore and aft orifice pin assemblies having 0.093 inch diameters. Two CCU-1/B cartridges shall be used in each of the above firings. Using one CCU-1/B, or CCU-9/B and one CCU-10/B cartridge, or equal, 50 releases of a 100 pound simulated store shall be performed with the fore and aft orifice pin assemblies having 0.156 inch diameters. The two ejector pistons and the slave piston shall be removed from the rack, disassembled and cleaned after each 25 firings or at the end of each firing day, whichever occurs first. The two ejector pistons shall retract completely for the first five firings following each cleaning of the rack. After each fiftieth firing, orifice pin assemblies, and all "O" rings in the slave and ejector piston assemblies shall be replaced. Completion of this test without such damage or erosion which precludes satisfaction of the functional test shall be considered as meeting the operational life requirements of this specification. Evidence of such damage, wear, or erosion shall be cause for rejection.

4.5.7 Environmental Tests. During or after, as applicable, each environmental test, the racks shall be subjected to steps f., g., and h. of the functional test. Failure to meet the requirements of these portions of the functional test shall be considered as failure to pass the environmental tests and cause for rejection.

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4.5.7.1 High Temperature. The racks shall be raised to and held at 176.7°C (350°F $^{+1}_{-0}$ percent for four hours. The temperature shall be reduced to ambient and the racks shall be fired five times as specified in the firing portion of the functional test and an inspection shall be made for gas leakage during the firings. In addition to the criteria in paragraph "4.5.7 Environmental Tests," evidence of gas leakage shall be cause for rejection.

4.5.7.2 Humidity. The racks shall be subjected to the humidity test described in procedure I, Method 507, MIL-STD-810.

4.5.7.3 Salt Fog. The rack shall be subjected to a salt fog test in accordance with MIL-T-7743. Paragraph 4.12 of MIL-T-7743 shall apply.

4.5.7.4 Icing Conditions. The rack shall be stabilized at 53.9°C (-65°F) $^{+0}_{-3}$ percent, then subjected to 37.7°C (100°F) $^{+3}_{-0}$ percent and 90 ± 2 percent relative humidity until frost has disappeared. The racks shall then be stabilized at -53.9°C (-65°F) $^{+0}_{-3}$ percent. While the racks are at this temperature, step f. of the functional test shall be performed. Force required for manual release shall not exceed 450 inch pounds.

4.5.7.5 Sand and Dust. The racks shall be subjected to a sand and dust test in accordance with procedure I, Method 510, MIL-STD-810.

4.5.7.6 Vibration. The racks shall be subjected to a vibration test in accordance with procedure I, Method 514, MIL-STD-810, except that figure 3 of this specification shall be utilized in lieu of figure 514-1 of MIL-STD-810. All electrical circuits shall be monitored during this test. The vibration test shall be performed once with the racks in each of the following configurations:

a. Rack, not loaded.

b. Loaded, safety lock locked.

   (1) With 250 pound simulated store installed.

   (2) With 1000 pound simulated store installed.

c. Loaded, safety lock unlocked.

   (1) With 250 pound simulated store installed.

   (2) With 1000 pound simulated store installed.

NOTE: The 250 pound simulated store shall have a minimum diameter of 10.75 inches.

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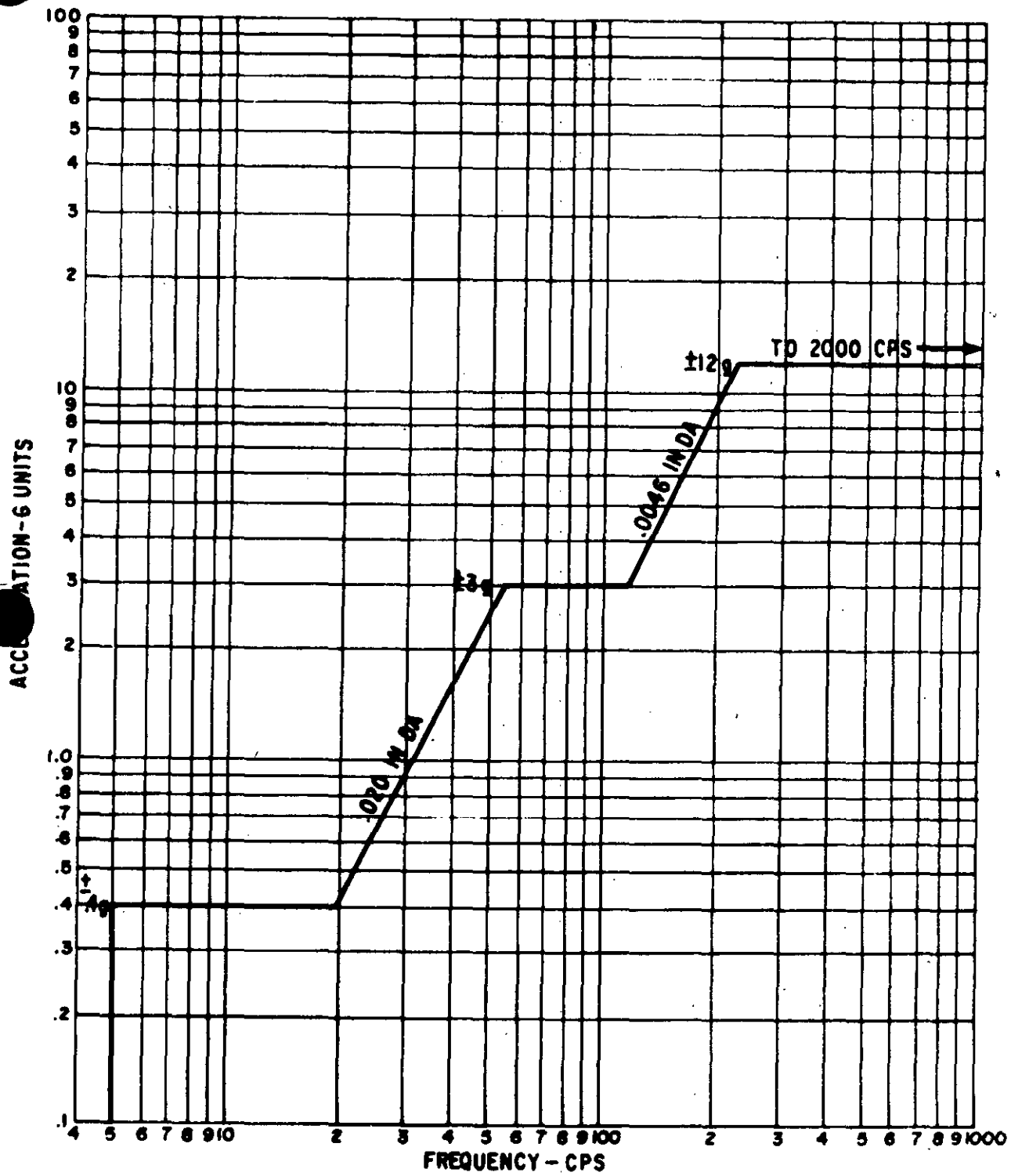


FIGURE - 3 RANGE CURVE FOR VIBRATION TEST

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In addition to the criteria of 4.5.7, the following shall be cause for rejection:

a. Malfunction of any part or circuit during the test.

b. Failure of any part during the test.

4.5.7.7 Shock. For this test, the maximum weight simulated store shall be the 2000 pound simulated store. Design loads shall be as illustrated in Table II. The rack shall be subjected to the yield load shock tests described in MIL-T-7743. This test shall be conducted with the inflight safety lock locked and repeated with the inflight safety lock unlocked.

4.5.8 Electrical Tests. After all preceding tests in 4.5 have been completed, a complete electrical continuity and insulation resistance test of all circuits as shown in Drawing No. 68J13136 shall be performed. The insulation resistance test shall be in accordance with MIL-STD-202, Method 302, Test Condition A. Insulation resistance shall be greater than 100 megohms. Failure to pass these tests shall be cause for rejection.

NOTE: After completion of the function firing, the rack shall be cleaned (see section 6).

5. PREPARATION FOR DELIVERY

5.1 Levels A. and B. Each rack shall be cleaned in accordance with process C-1 of MIL-P-116, and preserved in accordance with Method IIA of MIL-P-116. All rack projections, sharp edges, and components which might puncture or otherwise damage the barrier shall be padded or wrapped with PPP-C-843 cushioning material.

5.1.1 Level C. Each rack shall be cleaned in accordance with process C-1 of MIL-P-116 and preserved in accordance with Method III of MIL-P-116.

5.2 Packing

5.2.1 Levels A. and B. A number of racks, specified by the procuring activity, preserved and packaged in accordance with 5.1.1 shall be packed in containers conforming to MIL-B-26195, Style A, Type I. Cushioning, anchoring, bracing, blocking, and water-proofing shall be in accordance with MIL-STD-1186.

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5.2.2 Level C. Each rack, preserved and packaged in accordance with 5.1.2 shall be packed in containers so constructed to ensure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery.

5.3 Marking for Shipment and Storage. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

RACK, BOMB EJECTOR, AIRCRAFT BRU-4/A

or

RACK, BOMB EJECTOR, AIRCRAFT BRU-4A/A

6. NOTES

6.1 Intended Use. The BRU-4/A Bomb Rack and the BRU-4A/A Bomb Rack are universal racks designed to be attached to a variety of aircraft and will be used to carry and release special stores, conventional stores and fuel tanks.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Conditions under which Government loaned and Government furnished property will be made available to the contractor.

c. Required level of preservation, packaging and packing.

d. Number of spare "O" rings to be furnished with each rack.

6.3 Preproduction Samples. It is expected that the contract or purchase order will specify that a minimum of three units representative of each rack will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of inspection for these tests.

6.3.1 Sample Plan for Vendors. In the event multiple vendors of a component are utilized, or a change in vendors occurs, the contractor is obligated to provide one additional representative bomb rack unit for sampling tests containing said multiple vendor part or parts changed from approved qualification sample supplier without an increase in price.

6.4 Production Delivery. No production bomb rack should be delivered prior to full approval of the preproduction samples.

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6.5 Gages, Jigs, Fixtures, Dies, and Molds. All special gages, jigs, fixtures, dies or molds peculiar and necessary to the manufacture of the racks shall be provided as required by the contractor and shall become the property of the Government.

6.6 Salvaged Pieces. Any piece(s) from a rejected rack (exclusive of a piece(s) that has been subjected to loads in excess of the yield load point of the rack) may be salvaged for use providing each salvaged piece is inspected, and is found to be in complete compliance with the requirements of this specification and the pertinent drawings and specifications.

6.7 The racks shall be cleaned as required by the applicable Technical Order for daily cleaning before shipment. (Technical Order may be obtained from the procuring activity.)

6.8 Definitions

a. Preproduction Tests. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. Acceptance Tests. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

6.9 Differences of the Racks. The BRU-4A/A differs from the BRU-4/A in that 1/4 inch titanium side plates replace the 3/16 inch steel sides of the BRU-4/A providing a 5 pound weight savings and additional corrosion resistance.

Custodian
Air Force - 15

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EAC/to

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U.S. MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

AFWL (SEEE)
Kirtland AFB NM 87117

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to *preparing activity*. *Comments submitted on this form do not constitute or imply authorization to waive any* portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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